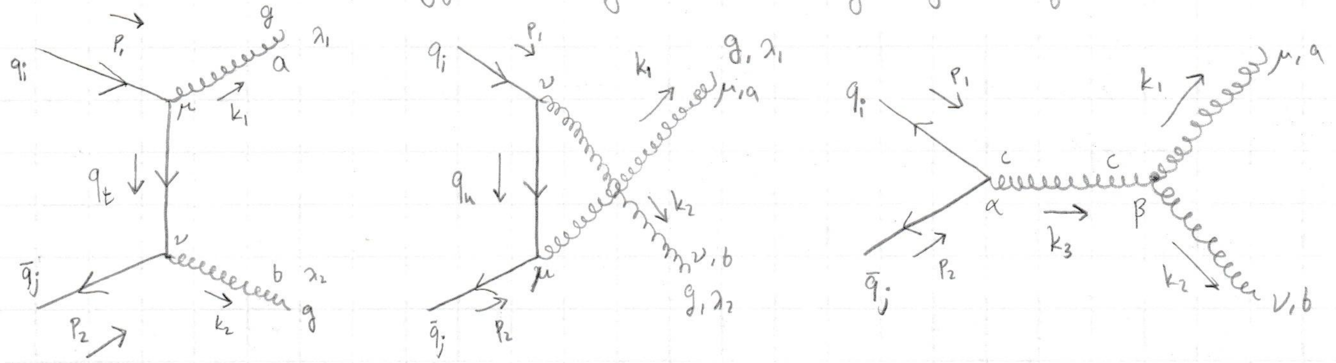
Quantum Field Theory II   Homework 4

Exercise 4.8

$\bar{q}q \to gg$ scattering has the following Feynman-diagrams:

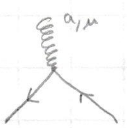


t-channel   u-channel   s-channel

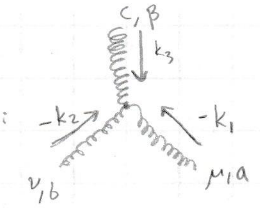
Here q_i is quark with colour i, $\bar{q}_j$ is antiquark with colour j. λ_1, λ_2 are polarization indices. μ, ν, α, β are gluon Lorentz indices and a, b, c are gluon group indices. We have chosen not to show spin since it is not relevant.

$$q_t = p_1 - k_1 = k_2 - P_2 \quad , \quad q_u = P_1 - k_2 = k_1 - P_2 \quad , \quad k_3 = P_1 + P_2 = k_1 + k_2$$

We use the following Feynman rules:

Fermion vertex: 

$= ig\gamma^\mu T^a$

Gluon-vertex: 

$$= g f^{abc}\left[g^{\mu\nu}(-k_1 + k_2)^\beta + g^{\nu\beta}(-k_2 - k_3)^\mu + g^{\beta\mu}(k_3 + k_1)^\nu \right]$$

Fermion propagator: $\underset{q}{\longleftarrow} = \dfrac{i}{\not{q} - m} = \dfrac{i}{\not{p}_1 - \not{k}_1 - m}$ for $q = P_1 - k_1$

External fermion: $\longrightarrow\!\bullet = u_i(p)$, where i is colour index.

Gluon propagator in Feynman-gauge: $\mu \underset{k_3}{\rightsquigarrow} \nu = -i \dfrac{g^{\mu\nu}}{k_3^2}$

External gluons: $\underset{k}{\leftarrow} = \varepsilon_\mu^\lambda(k)$, $\underset{k}{\rightarrow} = \varepsilon_\mu^{\lambda *}(k)$ for initial and final state respectively.

Using these we can compute the Amplitudes for $\mathcal{M}_{t+s}$ and $\mathcal{M}_s$

$$i\mathcal{M}_{t+u} = (ig)^2 \bar{v}_j(p_2) \Big\{ \underbrace{\gamma^\nu T^b \frac{i}{\not{p}_1 - \not{k}_1 - m} \gamma^\mu T^a}_{\text{t-channel part}} + \underbrace{\gamma^\mu T^a \frac{i}{\not{k}_1 - \not{p}_2 - m} \gamma^\nu T^b}_{\text{u-channel part}} \Big\} u_i(p_1)\, \varepsilon_\mu^{\lambda_1 *}(k_1)\, \varepsilon_\nu^{\lambda_2 *}(k_2)$$

$\boxed{i\mathcal{M}_{t+u} \equiv \mathcal{M}_{t+u}^{\mu\nu}\, \varepsilon_\mu^*(k_1, \lambda_1)\, \varepsilon_\nu^*(k_2, \lambda_2)}$, where we have defined $\varepsilon_\mu^*(k_1, \lambda_1) \equiv \varepsilon_\mu^{\lambda_1 *}(k_1)$ and

$$\mathcal{M}_{t+u}^{\mu\nu} \equiv (ig)^2 \bar{v}_j(p_2) \left\{ \gamma^\nu T^b \frac{i}{\not{p}_1 - \not{k}_1 - m} \gamma^\mu T^a + \gamma^\mu T^a \frac{i}{\not{k}_1 - \not{p}_2 - m} \gamma^\nu T^b \right\} u_i(p_1)$$

The s-channel amplitude is

$$i\mathcal{M}_s = ig\,\bar{v}_j(p_2)\gamma^\alpha T^c u_i(p_1)\,\frac{-ig_{\alpha\beta}}{k_3^2}\cdot g f^{abc}\left[g^{\mu\nu}(k_2-k_1)^\beta - g^{\nu\beta}(k_2+k_3)^\mu + g^{\beta\mu}(k_3+k_1)^\nu\right]\varepsilon_\mu^{\lambda_1 *}(k_1)\,\varepsilon_\nu^{\lambda_2 *}(k_2)$$

$$= -g^2\,\bar{v}_j(p_2)\gamma_\beta T^c u_i(p_1)\,\frac{f^{abc}}{k_3^2}\left(g^{\mu\nu}[k_2-k_1]^\beta - g^{\nu\beta}[k_2+k_3]^\mu + g^{\beta\mu}[k_3+k_1]^\nu\right)\varepsilon_\mu^*(k_1,\lambda_1)\,\varepsilon_\nu^*(k_2,\lambda_2)$$

$$= \mathcal{M}_s^{\mu\nu}\,\varepsilon_\mu^*(k_1,\lambda_1)\,\varepsilon_\nu^*(k_2,\lambda_2)$$

The total amplitude is $i\mathcal{M} = i\mathcal{M}_{t+u} + i\mathcal{M}_s$:

$$i\mathcal{M} = \Bigg[(ig)^2\,\bar{v}_j(p_2)\left\{\gamma^\nu T^b\frac{i}{\not p_1 - \not k_1 - m}\gamma^\mu T^a + \gamma^\mu T^a\frac{i}{\not k_1 - \not p_2 - m}\gamma^\nu T^b\right\}u_i(p_1) + g^2\,\bar{v}_j(p_2)\gamma_\beta T^c u_i(p_1)$$

$$\frac{f^{abc}}{k_3^2}\left[g^{\mu\nu}[k_2-k_1]^\beta - g^{\nu\beta}[k_2+k_3]^\mu + g^{\beta\mu}[k_3+k_1]^\nu\right]\Bigg]\,\varepsilon_\mu^*(k_1\lambda_1)\,\varepsilon_\nu^*(k_2,\lambda_2)$$

$$i\mathcal{M} \equiv \mathcal{M}^{\mu\nu}\,\varepsilon_\mu^*(k_1\lambda_1)\,\varepsilon_\nu^*(k_2\lambda_2)$$

Exercise 4.9

Show that $\sum_{\text{non-phys}} |\varepsilon_1^\mu \varepsilon_2^\nu \mathcal{M}_{\mu\nu}|^2 \neq 0$

The polarization sum for physical polarization is

$$\sum_{i=1}^{2}\varepsilon_\mu(k,i)\,\varepsilon_\nu^*(k,i) = -g_{\mu\nu} + \frac{\bar{k}_\mu k_\nu + k_\mu \bar{k}_\nu}{k\cdot\bar{k}} \equiv P_{\mu\nu}(k) \qquad (1)$$

The polarization sum for all states is

$$\sum_{\lambda=0}^{3}\varepsilon_\mu(k,\lambda)\,\varepsilon_\nu^*(k,\lambda) = -g_{\mu\nu} \qquad (2)$$

Using (1) we get $\sum_{\text{phys}}|\varepsilon_1^\mu\varepsilon_2^\nu\mathcal{M}_{\mu\nu}|^2 = \sum_{\lambda=0}^{2}(\mathcal{M}^{\mu\nu})^*\mathcal{M}^{\alpha\beta}\,\varepsilon_\mu(k_1,\lambda_1)\,\varepsilon_\nu(k_2,\lambda_2)\,\varepsilon_\alpha^*(k_1,\lambda_1)\,\varepsilon_\beta^*(k_2,\lambda_2)$

$$\Rightarrow \sum_{\text{phys}}|\varepsilon_1^\mu\varepsilon_2^\nu\mathcal{M}_{\mu\nu}|^2 = (\mathcal{M}^{\mu\nu})^*(\mathcal{M}^{\alpha\beta})\,P_{\mu\alpha}(k_1)\,P_{\nu\beta}(k_2)$$

And using (2) we get $\sum_{\text{all states}}|\varepsilon_1^\mu\varepsilon_2^\nu\mathcal{M}_{\mu\nu}|^2 = (\mathcal{M}^{\mu\nu})^*(\mathcal{M}^{\alpha\beta})(-g_{\mu\alpha})(-g_{\nu\beta})$

Using these expressions we get expression for non-physical sum:

$$\sum_{\text{non-phys}}|\varepsilon_1^\mu\varepsilon_2^\nu\mathcal{M}_{\mu\nu}|^2 = \sum_{\text{all states}}|\varepsilon_1^\mu\varepsilon_2^\nu\mathcal{M}_{\mu\nu}|^2 - \sum_{\text{phys}}|\varepsilon_1^\mu\varepsilon_2^\nu\mathcal{M}_{\mu\nu}|^2$$

$$= (\mathcal{M}^{\mu\nu})^*(\mathcal{M}^{\alpha\beta})\left[g_{\mu\alpha}g_{\nu\beta} - P_{\mu\alpha}(k_1)\,P_{\nu\alpha}(k_2)\right] \equiv S$$

$$\Rightarrow \quad S = (M^{\mu\nu})^* (M^{\alpha\beta}) \left[g_{\mu\alpha} g_{\nu\beta} - \left(-g_{\mu\alpha} + \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu}\bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1}\right)\left(-g_{\nu\beta} + \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu}\bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2}\right)\right]$$

$$S = (M^{\mu\nu})^* M^{\alpha\beta} \left[g_{\mu\alpha} \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu}\bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2} + g_{\nu\beta} \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu}\bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} - \frac{\bar{k}_{1\mu} k_{1\alpha} + k_{1\mu}\bar{k}_{1\alpha}}{k_1 \cdot \bar{k}_1} \frac{\bar{k}_{2\nu} k_{2\beta} + k_{2\nu}\bar{k}_{2\beta}}{k_2 \cdot \bar{k}_2}\right]$$

Now Let's calculate the contractions $M^{\mu\nu} k_{1\mu}$ and $M^{\mu\nu} k_{2\nu}$

$$k_{1\mu} M^{\mu\nu} = k_{1\mu} \left[M^{\mu\nu}_{t+u} + M^{\mu\nu}_{s} \right]$$, Let's start with $M^{\mu\nu}_{t+u} k_{1\mu}$

$$k_{1\mu} M^{\mu\nu}_{t+u} = i(ig)^2\, \bar{v}_j(p_2) \left\{ -\gamma^\nu T^b \frac{-\not{k}_1}{\not{p}_1 - \not{k}_1 - m} T^a + T^a \frac{\not{k}_1}{\not{k}_1 - \not{p}_2 - m} \gamma^\nu T^b \right\} u_i(p_1)$$

We now use Dirac equation : $(\not{p}_1 - m)\, u(p_1) = 0$ and $\bar{v}(p_2)(-\not{p}_2 - m) = 0$ to insert $(\not{p}_1 - m)$ and $(-\not{p}_2 - m)$ to the numerators via summation since $\not{k}_1 + 0 = \not{k}_1$

$$\Rightarrow \quad k_{1\mu} M^{\mu\nu}_{t+u} = i(ig)^2\, \bar{v}_j(p_2) \left\{ -\gamma^\nu T^b \underbrace{\frac{\not{p}_1 - \not{k}_1 - m}{\not{p}_1 - \not{k}_1 - m}}_{=1} T^a + T^a \underbrace{\frac{\not{k}_1 - \not{p}_2 - m}{\not{k}_1 - \not{p}_2 - m}}_{=1} \gamma^\nu T^b \right\} u_i(p_1)$$

$$k_{1\mu} M^{\mu\nu}_{t+u} = i(ig)^2\, \bar{v}_j(p_2) \left\{ -\gamma^\nu T^b T^a + T^a T^b \gamma^\nu \right\} u_i(p_1) \qquad \| \quad T^a T^b - T^b T^a = [T^a, T^b] = i f^{abc} T^c$$

$$= (ig)^2\, \bar{v}_j(p_2) \left\{ i^2 f^{abc} T^c \gamma^\nu \right\} u_i(p_1) \qquad \| \text{ Colour : } \bar{v}_j = \underset{\text{spinor}}{\bar{v}} \otimes \underset{\text{colour space basis vector}}{\hat{c}_j}$$

$$k_{1\mu} M^{\mu\nu}_{t+u} = f^{abc} g^2\, \bar{v}(p_2) \gamma^\nu u(p_1)\, T^c_{ji} \qquad \text{and } c_j T^a c_i = T^a{}_{ij}$$

Now for the s-channel term

$$k_{1\mu} M^{\mu\nu}_{s} = g^2\, \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} k_{1\mu} \left[g^{\mu\nu}(k_2 - k_1)^\beta - g^{\nu\beta}(k_2 + k_3)^\mu + g^{\beta\mu}(k_3 + k_1)^\nu \right]$$

$$= g^2\, \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} \left[k_1^\nu (k_2 - k_1)^\beta - g^{\nu\beta}(k_2 + k_3)\cdot k_1 + k_1^\beta (k_3 + k_1)^\nu \right]$$

$$(k_2 + k_3)\cdot k_1 = (k_2 + k_3)(k_3 - k_2) = \overset{=0}{k_2^2} + k_3^2 = k_3^2 \qquad \| \; k_2^2 = 0 \text{ for on-shell gluon}$$

$$k_1^\nu (k_2 - k_1)^\beta + k_1^\beta (k_3 + k_1)^\nu = k_1^\nu k_2^\beta - k_1^\nu k_1^\beta + k_1^\beta k_3^\nu + k_1^\beta k_1^\nu = k_1^\nu k_2^\beta + k_1^\beta k_3^\nu \qquad \| \; k_1^\nu = (k_3 - k_2)^\nu$$

$$= k_3^\nu (k_2 + k_1)^\beta - k_2^\nu k_2^\beta = k_3^\nu k_3^\beta - k_2^\nu k_2^\beta$$

$$\Rightarrow \quad k_{1\mu} M^{\mu\nu}_{s} = g^2\, \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} \left[k_3^\nu k_3^\beta - k_2^\nu k_2^\beta - g^{\nu\beta} k_3^2 \right]$$

The term $k_3^\beta k_3^\nu$ vanishes when contracted with the fermion current so we are left with

$$k_{1\mu} M^{\mu\nu}_{s} = -g^2\, \bar{v}(p_2) \gamma^\nu u(p_1)\, T^c_{ji} f^{abc} - k_2^\nu g^2\, \bar{v}(p_2) \not{k}_2 u(p_1)\, T^c_{ji} \frac{f^{abc}}{k_3^2}$$

Now we add the result and get:

$$k_{1\mu} M^{\mu\nu} = k_{1\mu} M^{\mu\nu}_{t+u} + k_{1\mu} M^{\mu\nu}_{s} = -k_2^{\nu} \frac{g^2 f^{abc}}{k_3^2} T^c_{ji} \bar{v}(p_2) \not{k}_2 u(p_1) \qquad \Big\| \; k_3^2 = k_1^2 + k_2^2 + 2k_1 \cdot k_2$$

$$k_{1\mu} M^{\mu\nu} = -k_2^{\nu} \frac{g^2 f^{abc}}{2\, k_1 \cdot k_2} T^c_{ji} \bar{v}(p_2) \not{k}_2 u(p_1) \equiv k_2^{\nu} A(k_1)$$

Similarily we get

$$k_{2\nu} M^{\mu\nu}_{t+u} = i(ig)^2 \bar{v}_j(p_2) \left\{ T^b \frac{\not{k}_2}{\not{k}_2 - \not{p}_2 - m} \gamma^\mu T^a + \gamma^\mu T^a \frac{\not{k}_2}{\not{p}_1 - \not{k}_2 - m} T^b \right\} u(p_1)$$

We have used momentum conservation to get the denominators in terms of k_2

t-channel : $p_1 - k_1 = k_2 - p_2$, u-channel : $p_1 - k_2 = k_1 - p_2$

Now we use Dirac equation like previously :

$$k_{2\nu} M^{\mu\nu}_{t+u} = i(ig)^2 \bar{v}_j(p_2) \left\{ T^b \gamma^\mu T^a - \gamma^\mu T^a T^b \right\} u(p_1)$$

$$= i(ig)^2 \bar{v}_j(p_2) \left\{ -i f^{abc} T^c \gamma^\mu \right\} u(p_1) = -g^2 f^{abc} T^c_{ji} \bar{v}(p_2) \gamma^\mu u(p_1)$$

And for the s-channel

$$k_{2\nu} M^{\mu\nu}_{s} = g^2 \bar{v}_j(p_2) \gamma_\beta T^c u_i(p_1) \frac{f^{abc}}{k_3^2} \left(g^{\mu\nu}(k_2 - k_1)^\beta - g^{\nu\beta}(k_2 + k_3)^\mu + g^{\beta\mu}(k_3 + k_1)^\nu \right) k_{2\nu}$$

$$= g^2 T^c_{ji} \bar{v}(p_2) \gamma_\beta u(p_1) \frac{f^{abc}}{k_3^2} \left(k_2^\mu (k_2 - k_1)^\beta - k_2^\beta (k_2 + k_3)^\mu + g^{\beta\mu} (k_3 + k_1) \cdot k_2 \right)$$

$$(k_3 + k_1) \cdot k_2 = (k_3 + k_1)(k_3 - k_1) = k_3^2 \;, \qquad k_2^\mu (k_2 - k_1)^\beta - k_2^\beta (k_2 + k_3)^\mu = -k_2^\mu k_1^\beta - k_2^\beta k_3^\mu$$

$$= -(k_3^\mu - k_1^\mu) k_1^\beta - k_2^\beta k_3^\mu = -k_3^\mu k_3^\beta - k_1^\mu k_1^\beta \qquad \|$$

$$\Rightarrow \quad k_{2\nu} M^{\mu\nu}_{s} = g^2 T^c_{ji} \bar{v}(p_2) \gamma^\mu u(p_1) f^{abc} - g^2 T^c_{ji} f^{abc} \bar{v}(p_2) \gamma_\beta u(p_1) \left[k_3^\mu k_3^\beta + k_1^\mu k_1^\beta \right] \cdot \frac{1}{k_3^2}$$

As before $k_3^\mu k_3^\beta$ vanishes when contracted by fermion current and $k_3^2 = 2 k_1 \cdot k_2$

$$\Rightarrow \quad k_{2\nu} M^{\mu\nu} = k_{2\nu} M^{\mu\nu}_{t+u} + k_{2\nu} M^{\mu\nu}_{s} = -k_1^\mu \frac{g^2 f^{abc}}{2 k_1 \cdot k_2} T^c_{ji} \bar{v}(p_2) \not{k}_1 u(p_1)$$

$$k_{2\nu} M^{\mu\nu} \equiv k_1^\mu A(k_2)$$

$T^{c\dagger} = T^c \Rightarrow T^{c\dagger}_{ij} = T^c_{ij}$ ✓

We will also use relation $k_{2\nu} (M^{\mu\nu})^* = k_1^\mu A^* = -k_1^\mu \frac{g^2 f^{abc}}{2 k_1 \cdot k_2} T^c_{ji} \bar{u}(p_1) \not{k}_1 v(p_2)$

since $(M^{\mu\nu})^*$ affects only spinor and colour part of which we take Hermitean adjoint of

$\bar{k}_{2\nu} = R\, k_{2\nu} M^{\mu\nu} = R\, k_1^\mu A = \bar{k}_1^\mu A$, where R is operator that turns $k \to \bar{k}$.

$$\Rightarrow \quad \underline{\bar{k}_{2\nu} = \bar{k}_1^\mu A(k_2)}$$

Now we use the Relations $k_{1\mu}\mathcal{M}^{\mu\nu} = k_2^{\nu} A(k_1)$ etc to simplify our S:

$$S = (\mathcal{M}^{\mu\nu})^* \mathcal{M}^{\alpha\beta}\left[g_{\mu\alpha}\frac{\bar k_{2\nu}k_{2\beta}+k_{2\nu}\bar k_{2\beta}}{k_2\cdot\bar k_2} + g_{\nu\beta}\frac{\bar k_{1\mu}k_{1\alpha}+k_{1\mu}\bar k_{1\alpha}}{k_1\cdot\bar k_1} - \left(\frac{\bar k_{1\mu}k_{1\alpha}+k_{1\mu}\bar k_{1\alpha}}{k_1\cdot\bar k_1}\right)\cdot\left(\frac{\bar k_{2\nu}k_{2\beta}+k_{2\nu}\bar k_{2\beta}}{k_2\cdot\bar k_2}\right)\right]$$

Let's calculate term by term

$$\frac{1}{k_2\cdot\bar k_2}(\mathcal{M}^{\mu\nu})^*\mathcal{M}^{\alpha\beta} g_{\mu\alpha}\left(\bar k_{2\nu}k_{2\beta}+k_{2\nu}\bar k_{2\beta}\right) = \frac{g_{\mu\alpha}}{k_2\cdot\bar k_2}\left[\bar k_1^{\mu}A^*(k_2)\,k_{2\beta} + k_1^{\mu}A^*(k_2)\,\bar k_{2\beta}\right]\mathcal{M}^{\alpha\beta}$$

$$= \frac{1}{k_1\cdot\bar k_2}\left[\bar k_{1\alpha}k_{2\beta}A^*(k_2) + k_{1\alpha}\bar k_{2\beta}A^*(k_2)\right]\mathcal{M}^{\alpha\beta}$$

$$= \frac{1}{k_2\cdot\bar k_2}\underbrace{\left[\bar k_2^{\beta}k_{2\beta} + k_2^{\beta}\bar k_{2\beta}\right]}_{=2k_2\cdot\bar k_2}A(k_1)A^*(k_2) = 2A(k_1)A^*(k_2)$$

Similarity we get

$$\frac{1}{k_1\cdot\bar k_2}(\mathcal{M}^{\mu\nu})^*\mathcal{M}^{\alpha\beta}\left[\bar k_{1\mu}k_{1\alpha}+k_{1\mu}\bar k_{1\alpha}\right] = 2A(k_2)A^*(k_1)$$

And

because $\bar k^2 = 0$, $k^2 = 0$

$$(\mathcal{M}^{\mu\nu})^*\mathcal{M}^{\alpha\beta}\left[\frac{\bar k_{1\mu}k_{1\alpha}+k_{1\mu}\bar k_{1\alpha}}{k_1\cdot\bar k_1}\right]\left[\frac{\bar k_{2\nu}k_{2\beta}+k_{2\nu}\bar k_{2\beta}}{k_2\cdot\bar k_2}\right] = \frac{(\mathcal{M}^{\mu\nu})^*\mathcal{M}^{\alpha\beta}}{(k_1\cdot\bar k_1)(k_2\cdot\bar k_2)}\Big[\overbrace{\bar k_{1\mu}k_{1\alpha}\bar k_{2\nu}k_{2\beta}}^{=\text{ gives } 0} + \bar k_{1\mu}k_{1\alpha}k_{2\nu}\bar k_{2\beta} + k_{1\mu}\bar k_{1\alpha}\bar k_{2\nu}k_{2\beta} + \underbrace{k_{1\mu}\bar k_{1\alpha}k_{2\nu}\bar k_{2\beta}}_{=0}\Big]$$

$$= \frac{1}{(k_1\cdot\bar k_1)(k_2\cdot\bar k_2)}\left[\bar k_2^{\nu}A^*(k_1)\,k_{1\alpha}\,k_{2\nu}\,\bar k_1^{\beta}A(k_2) + k_{1\mu}\bar k_2^{\beta}A(k_1)\,\bar k_{1\mu}A^*(k_2)\,k_{2\beta}\right]$$

$k_1\cdot\bar k_1$ $\quad$ $k_2\cdot\bar k_2$

$$= A(k_2)A^*(k_1) + A(k_1)A^*(k_2)$$

$$\Rightarrow \quad S = 2A(k_1)A^*(k_2) + 2A(k_2)A^*(k_1) - \left(A(k_1)A^*(k_2) + A(k_2)A^*(k_1)\right)$$

$$S = A(k_1)A^*(k_2) + A(k_2)A^*(k_1)$$, Now that we have simplified this

let's expand again : with $A(k_1) = -\frac{g^2 f^{abc}}{2k_1\cdot k_2}T^c_{ji}\,\bar v(p_2)\not{k}_2\,u(p_1)$, $A(k_2) = -\frac{g^2 f^{abc}}{2k_1\cdot k_2}T^c_{ji}\,\bar v(p_2)\not{k}_1\,u(p_1)$

and $A^*(k_1) = -\frac{g^2 f^{abc}}{2k_1\cdot k_2}T^c_{ji}\,\bar u(p_1)\not{k}_2\,v(p_2)$, $A^*(k_2) = -\frac{g^2 f^{abc}}{2k_1\cdot k_2}T^c_{ji}\,\bar u(p_1)\not{k}_1\,v(p_1)$

$$\Rightarrow \quad A(k_1)A^*(k_2) = \frac{g^4 f^{abc}}{4(k_1\cdot k_2)^2}f^{mno}(T^c_{ji})(T^o_{ji})\,\bar v(p_2)\not{k}_2\,u(p_1)\,\bar u(p_1)\not{k}_1\,v(p_2)$$

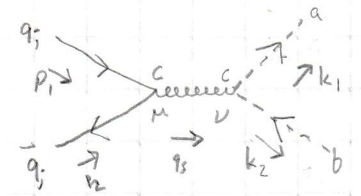
$$\Rightarrow \quad A(k_2)A^*(k_1) = \frac{g^4 f^{abc}}{4(k_1\cdot k_2)^2}f^{mno}(T^c_{ji})(T^o_{ji})\,\bar v(p_2)\not{k}_1\,u(p_1)\,\bar u(p_1)\not{k}_2\,v(p_2)$$

$$\Rightarrow \quad S = -\frac{g^4 f^{abc}}{4(k_1\cdot k_2)^2} f^{mno} (T^c)_{ji} (T^o)_{ji}\, \bar{v}(p_2) \left[\not{k}_2 u(p_1)\bar{u}(p_1) \not{k}_1 + \not{k}_1 u(p_1)\bar{u}(p_1) \not{k}_2 \right] v(p_2) \neq 0$$

$\Rightarrow$ We have shown that $\sum_{\text{non-phys}} |\varepsilon_1^\mu \varepsilon_2^\nu M_{\mu\nu}|^2 \neq 0$

$\boxed{\text{Exercise 4.10}}$

For $q\bar{q} \to c\bar{c}$ we only have s-diagram since there is no direct interaction / coupling of q and c fields. The Feynman diagram is then

$\Rightarrow$ 

The ghost vertex gives the contribution: $-g f^{acb} k_1^\nu = g f^{abc} k_1^\nu$ since f^{abc} is totally antisymmetric

We note that the vertex depends on the momentum of ghost-particle so we actually get two contribution from the s-channel: one where ghost has momentum of k_2 and one where it has momentum of k_1. Because of this our Matrix element looks like the following: (External legs of scalar field is 1)

$$i M_{\text{ghost}} = \bar{v}_j(p_2) \left[+ig\gamma^\mu T^c \right] u_i(p_1) \left(-\frac{i g_{\mu\nu}}{q_s^2} \right) \cdot \left(g f^{abc} k_1^\nu + g f^{abc} k_2^\nu \right) \qquad \Big\| \quad q_s^2 = (k_1+k_2)^2 = -2k_1\cdot k_2$$

two different contributions from two different possible final state

$$= -\frac{g^2}{2(k_1\cdot k_2)} f^{abc} T^c_{ji}\, \bar{v}(p_2 \left[\not{k}_1 + \not{k}_2 \right] u(p_1)$$

$$\Rightarrow \quad |M_{\text{ghost}}|^2 = \frac{g^4}{4(k_1\cdot k_2)^2} f^{abc} f^{mno} T^c_{ji} T^o_{ji} \left(\bar{v}(p_2)(\not{k}_1+\not{k}_2)u(p_1) \right)\left(\bar{u}(p_1)\left[\not{k}_1+\not{k}_2\right]v(p_2) \right)$$

$$= \frac{g^4}{4(k_1\cdot k_2)^2} f^{abc} f^{mno} T^c_{ji} T^o_{ji}\; \bar{v}(p_2)\not{k}_1 u(p_1)\bar{u}(p_1)\left[\not{k}_1+\not{k}_2\right]v(p_2) + \bar{v}(p_2)\not{k}_2 u(p_1)\bar{u}(p_1)\left[\not{k}_1+\not{k}_2\right]v(p_2)$$

The terms $\bar{v}(p_2)\not{k}_1 u(p_1)\bar{u}(p_1)\not{k}_1 v(p_2)$ will vanish because it will be proportional to $k_1^2 = 0$ when taking the trace so we set it 0, same with $\not{k}_1$ replaced by $\not{k}_2$

$$\Rightarrow \quad |M_{\text{ghost}}|^2 = \frac{g^4}{4(k_1\cdot k_2)^2} f^{abc} f^{mno} T^c_{ji} T^o_{ji}\; \bar{v}(p_2)\left[\not{k}_2 u(p_1)\bar{u}(p_1)\not{k}_1 + \not{k}_1 u(p_1)\bar{u}(p_1)\not{k}_2 \right] v(p_2)$$

This term is exactly same ase $S = \sum_{\text{non-phys}} |\varepsilon_1^\mu \varepsilon_2^\nu M_{\mu\nu}|^2$ but with opposite sign

$$\Rightarrow \quad \sum_{\text{non-phys}} |\varepsilon_1^\mu \varepsilon_2^\nu M_{\mu\nu}|^2 + |M_{\text{ghost}}|^2 = 0 \qquad \square$$

So $$\sum_{\text{all}} |\varepsilon_1^\mu \varepsilon_2^\nu M_{\mu\nu}|^2 + |M_{\text{ghost}}|^2 = \sum_{\text{physica}} |\varepsilon_1^\mu \varepsilon_2^\nu M_{\mu\nu}|^2$$